KARAN KAPUR

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I build agent infrastructure and agentic tools for GTM teams: runtimes, human approval gates, and checks that show when an agent's output can be trusted. Most recently I shipped a scheduling app for the Wisconsin Union and co-founded thaw, an open-source inference library. Looking for a full-time role in applied AI or AI infrastructure.

EDUCATION

University of Wisconsin-Madison, Madison, WI May 2026
Bachelor of Science in Computer Science and Data Science | Minor in Entrepreneurship
OpenAI Open Source Contributor Fellowship | SAIL x OpenAI Fellow | First place, Transcend UW | 3x solo founder

PROJECTS

Foreman *(open-source agent runtime and workflow platform)* code
Built solo: a platform where a plain-English job becomes a versioned AI agent. **Per-call human approval** gates anything leaving the workspace, **spend caps** limit model calls, and deliverables are scored by checks, an LLM judge and feedback. A **leased Postgres queue with heartbeats** recovers crashed runs; ten crash and race cases tested. Used by 25+ clients.

RelayIQ *(lead enrichment and CRM agent)* code
Built solo: an agent that decides which leads deserve enrichment spend, which provider answers each field and whether to write to the CRM, with idempotent writes and human escalation. On a synthetic 495-case benchmark, **static field-level routing cut cost per usable lead from 13.24 to 4.07 credits**; learned routing did worse. Its confidence score, tested on 783 synthetic fields, had **Brier 0.164 and ECE 0.091** and was overconfident above 0.8. Used by 100+ people.

tell *(buying signals for sales teams)* code
Built solo: a pipeline that turns job boards, SEC filings and news into scored signals with a source link behind every score, with an MCP server and HubSpot write-back. **40+ clients and 5+ salespeople** use it. Hand-checking 41 production SEC filing links found 20 right, 20 wrong, 1 unclear; guards replayed on a production copy keep the 20 and send 21 to review.

thaw *(open-source LLM inference)* code
Co-founded with N. Matteson and M. Yu (Rust, CUDA, vLLM); I own the storage layer for model weights and KV-cache snapshots. The team's warm-pool fork round (four branches, H100 80GB, Llama-3.1-8B) has a **0.88s median**.

PROFESSIONAL EXPERIENCE

Wisconsin Union, Madison, WI Aug 2026 to present
Data Analyst, Application Development.
- Replaced the team's legacy software in 2 days with a production full-stack app for concurrent shift scheduling and work chat that **250+ employees** use to run **4,400+ shifts**, and owned it end to end from self-serve accounts to deployment.

Eurofins Scientific, Madison, WI Jan 2026 to May 2026
Software Engineer Intern.
- Interviewed lab technicians across **4+ business units** to turn their workflows into requirements, then took the tool from discovery to production, where technicians still use it.
- Turned **100K+** messy lab records into forecasts with confidence intervals: Python ETL, PostgreSQL, FastAPI, React, TypeScript.

OpenAI x UW-Madison SAIL Program, Madison, WI Jun 2025 to Aug 2025
Software Engineer.
- Built a GPT-4 RAG platform for UW-Madison IT (LangChain, Pinecone, C# WebSockets) that cut manual lookup **40%** and cloud cost **25%** for **60+ users**, at sub-2s latency for 50+ concurrent queries.

UW-Madison, Dept. of Electrical & Computer Engineering, Madison, WI Jan 2025 to May 2026
Computer Support Assistant.
- Resolved 1,000+ user issues at 98% satisfaction across Windows, macOS and Linux and cut downtime 40% with PowerShell log analysis across 1,500+ computers and servers.

Earlier: Automation and Compliance Engineering Intern, HCLSoftware (2024)

TECHNICAL SKILLS & CERTIFICATIONS

Agents and evals: LLM agents and tool calling, human approval gates, LLM-judge evals, confidence evaluation (Brier, ECE), RAG, MCP servers, Claude Code and Codex daily
Languages and platforms: Python, TypeScript, Rust, SQL; FastAPI, Next.js, React, PostgreSQL, Docker, Snowflake, AWS, Azure, GitHub Actions, PyTorch, Linux
Customer and GTM work: customer discovery, HubSpot and CRM integrations, webhooks, REST APIs
Certifications: AWS Certified Cloud Practitioner (2026); Anthropic: Model Context Protocol (Introduction and Advanced), Building with the Claude API (2026); IBM Machine Learning Professional Certificate (2025)